

Contemporary C++: *Learning Modern C++ in a Modern Way*



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Agenda 21/30

Session 21. Default constructors, Const member functions, and More

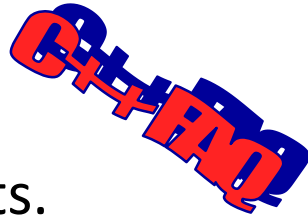
- Default ctors
- Examining State: Const member functions
- The size of C structs and C++ classes
- References and further readings
- Q&A

120 min (incl. Q & A)



Default constructor

- A **default constructor**, is a constructor that *can* be called without supplying an argument.
- If a user has declared a default constructor, that one will be used.



Q *Is the default constructor for Fred always Fred::Fred()?*

A No. A "default constructor" is a constructor that *can be called* with no arguments. One example of this is a constructor that takes no parameters:

```
class Fred {  
public:  
    Fred();    // Default constructor: can be called with no args  
    ...  
};
```

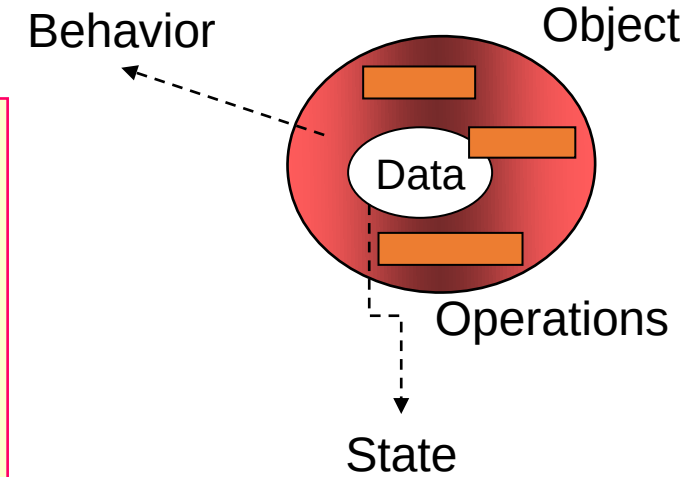
Another example of a "default constructor" is one that can take arguments, provided they are given default values:

```
class Fred {  
public:  
    Fred(int i=3, int j=5);    // Default constructor: can be called with no args  
    ...  
};
```

Examining State: Const member functions

- No modifying member functions

```
class Date {  
    int Day, Month, Year;  
public:  
    int day(); // note: "day" not "Day"  
    int month();  
    int year();  
};  
int Date::day() { return Day; }  
int Date::month() { return Month++; } // change state by mistake
```



- Constant objects

```
const char blank = ' ';  
blank = '\\0'; // error
```

```
const Date my_birthday(21, 7, 1971);
```

- Constant functions do not modify the state of objects of a given *type*.

```
class Date {  
    int Day, Month, Year;  
public:  
    // functions for examining the Date (Selectors)  
    int day() const { return Day; }  
    int month() const { return Month; }  
    int year() const; // just declaration  
};
```

Const member functions cont.

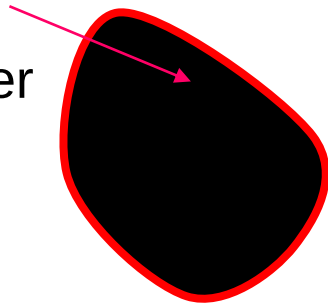
```
inline int Date::year() const {  
    return ++Year; // error: attempt to change member value in const function  
}
```

- When a const member function is defined outside its class, the const suffix is required.

```
inline int Date::year() {  
    return Year; // error: const missing in member function definition  
}
```

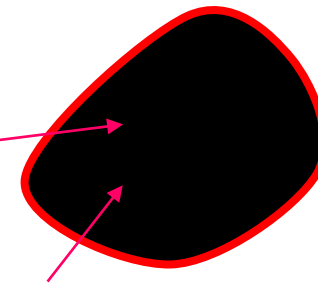
- Constant objects

constant member
functions



- Non-constant objects

Non-constant
member
functions



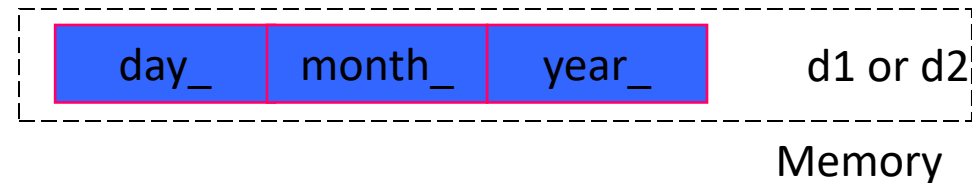
constant member
functions

Classes without Virtual functions: data members

- ... I always maintained a clear view of what an object looked like in memory.
 - Bjarne Stroustrup
- A class without a virtual function is a simple C struct.
- Each class object maintains its copy of the class data members.

```
class Date {  
    int day_, month_, year_;  
public:  
    Date(int = 0, int = 0, int = 0);  
  
    void add_year(int);  
    void add_month(int);  
    void add_day(int);  
    // Other member function(s)  
} d1;
```

```
struct Date {  
    int day_, month_, year_;  
} d2;
```

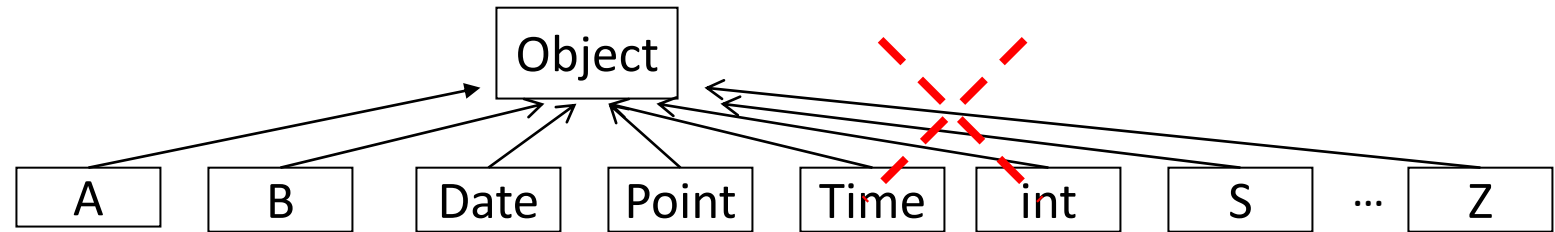


- Later members have higher addresses within a class object.

- A class without a virtual function requires exactly as much space to represent as a struct with the same data members.
- A compiler may add some “padding” between and after the members for alignment.

Classes without Virtual functions: member functions

- There is no generic/universal root class like Object!



- There exists only one instance of each class member function.
- C++ compiler uses the this pointer.

1

```
struct S {  
    int a;  
    int f(int) const;  
    int g() { return a++; }  
    int h() const { return a++; } // error  
};  
int S::f(int i) const { return a + i; }
```

2

```
void user()  
{  
    S s;  
    int i = s.f(2);  
    int j = s.g();  
}
```

translates to a global *decorated* function

```
X* const this; // object internal
```

```
__S__f(&s, 2);
```

this pointer

```
__S__g(&s);
```

this pointer

- non-static const member function:

```
const X* const this; // object internal
```

R eferences and further readings

[Str13] Bjarne Stroustrup. The C++ Programming Language. 4th edition. Addison Wesley. Reading Mass. USA. May 2013.

Thanks for your patience ...

A man who asks a question is a fool for minute,
The man who does not ask, is a fool for a life.
- Confucius

Learning to ask the right (often hard) questions is an essential part of learning to think as a programmer.

- Bjarne Stroustrup *programming Principles and Practice Using C++, page 4.*

There is no stupid question, but there is stupid answer.
- Howard Hinnant

